



United International University (UIU)

Midterm Examination

IPE 401/IPE3401: Industrial Management/Industrial & Operational Management

Spring Trimester: 2026

Total time: 1.5 hours

Total marks: 30

There are 3 questions. You must answer all questions

1. (a) Mr. Apurba plans to reach a retirement goal of \$2,000,000 by investing in an account with an 18% annual interest rate compounded monthly over 25 years. Calculate his required initial investment. Then, determine the time (in years) required for this initial investment to grow to a milestone of \$500,000 under the same interest conditions. (2.5) [CO1]
1. (b) A local entrepreneur is evaluating a three-year project. The interest rate is 12% compounded weekly. (4.5) [CO1]

Year	0	1	2
Cash Inflow (\$)	0	x	5000
Cash Outflow (\$)	2000	1000	y

The Future Value (FV) of all net cash flows at the end of Year 2 is \$4,000.

If $x + y = 3500$, Calculate the values of x and y .

1. (c) A manufacturing firm is considering investing in a new automated system. The cash flows for the investment are: (03) [CO1]

Year	0	1	2	3	4	5
Cash Flow (\$)	-32200	8500	11000	12080	9000	7000

If the rate of return is 12% per year, then determine the discounted payback period.

Should the project be accepted if the acceptable payback period is 4 years?

2. (a) The cash flows for two independent projects, Project A and Project B, are given below (05) [CO1]

Project A:

Year	0	1	2	3	4	5
Cash Flow (\$)	-40000	12000	15000	16000	10000	8000

Project B:

Year	0	1	2	3	4	5
Cash Flow (\$)	-40000	8000	10000	12000	14000	15000

Calculate IRR for both projects, starting from 12%, using the trial-and-error method. If the required rate of return (MARR) is 15%, which project(s) should be accepted?

2. (b) A popular energy drink, *Aura*, incurs a monthly total fixed cost of \$36,000 for plant operations and administrative expenses. Each bottle is sold at a unit selling price of \$120, while the unit variable cost is \$80 for ingredients, packaging, and labor. (2+2+1) [CO2]
- Find the break-even point in units.
 - Calculate the average fixed costs and average variable costs at the break-even level of output.
 - Determine the total cost if 1,200 units of bottled juice are sold in a month.

3. (a) A fan manufacturer uses 4,000 blades per month for its popular Jhor Utheche model. The blades are manufactured internally at a rate of 800 units per day. The most recent cost analysis shows that the annual holding cost for a unit of blade is \$1. If the setup cost is \$45 per production run and the firm operates 240 days per year, calculate the maximum inventory level and the annual setup cost, assuming the firm's batch size exactly matches its EPQ. (3) [CO2]

3. (b) A retailer sources shirts from a supplier who offers the following schedule: (07) [CO2]

Discount Number	Order Quantity	Discount (%)	Discount Price (\$)
1	0-499	0	25
2	500-999	12	?
3	1000+	20	?

The retailer sells 10,000 shirts annually. The cost of ordering shirts each time is \$75, while the holding cost is 20% of the unit price. Determine the **economic order quantity (EOQ)** and **associated unit cost**.

CO1-Apply Engineering economics and simple mathematics for solving project selection problems for choosing the best possible project.

CO2- Analyze various industrial problems by using operations management techniques, operations research techniques, and cost accounting techniques, and solve them.